KUMKUM RATHEE

B.Tech Information Technology • Full-Stack Developer • Competitive Programmer

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ABOUT ME / SUMMARY

B.Tech Information Technology student at BPIT maintaining a **9.5 CGPA** with strong foundations in C++, Data Structures & Algorithms, and Object-Oriented Programming. Solved **400+ coding problems**, published a peer-reviewed AI/ML research paper in **Taylor & Francis**, and earned an **NPTEL Elite + Silver** certification in Machine Learning (83%). 2x Smart India Hackathon internal round qualifier building robust, scalable full-stack web applications focused on real-time data, monitoring, and user-centered design.

EDUCATION

Bhagwan Parshuram Institute of Technology (BPIT), GGSIPU — 2024 – 2028
Bachelor of Technology (B.Tech) in Information Technology — **CGPA: 9.5 / 10.0**

Central Board of Secondary Education (CBSE) — 2024
Class XII (Senior Secondary) • Science Stream — **90.0%**

Central Board of Secondary Education (CBSE) — 2022
Class X (Secondary School) — **97.4%**

TECHNICAL SKILLS

Languages:	C++, C, Python, JavaScript (ES6+), SQL, HTML5, CSS3
CS Fundamentals:	Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks, Multithreading
Web & Backend:	Node.js, Express.js, RESTful APIs, Firebase (Firestore & Auth), JWT, Leaflet.js, Responsive UI
Tools & Databases:	Git, GitHub, VS Code, MySQL, SQLite, Postman, Vite

PROJECTS

MineGov AI – Mining Operations & Safety Platform [Source Code]

JavaScript (ES6+), Vite, Leaflet.js, AI Analytics • *Smart India Hackathon (SIH) Qualifier*

- Architected 4-tier hierarchical access gateway (*Coal India → Subsidiary → Area → Mine*) with **5 distinct role-based dashboards** for safety officers and executives.
- Integrated AI risk scoring, real-time alert severity categorization, automated SOS emergency dispatch, and live satellite fleet tracking on interactive Leaflet maps.

SmartSync – Logistics & Supply Chain Management Platform [Live Demo] • [Source Code]

JavaScript (ES6+), Firebase Auth, Cloud Firestore, Google Maps API, CSS3

- Engineered full-stack logistics portal supporting Firebase Email/Password Auth plus instant offline localStorage demo mode.
- Developed visual 5-stage shipment lifecycle tracker (*Booked → Dispatched → In Transit → Out for Delivery → Delivered*), slot reservation calendar, and automated route distance calculation.

SmartTour – AI-Assisted Travel Planning Suite [Live Demo] • [Source Code]

Node.js, Express.js, JWT, Bcrypt, Leaflet.js, REST APIs

- Built 4-step itinerary engine with dynamic REST API `/api/ai/generate` generating day-wise customized schedules.
- Implemented secure token authentication, encrypted credentials, cloud sync, and interactive drag-and-drop itinerary studio.

Secure Multi-Room TCP Socket Chat Application [Source Code]

Python, TCP Sockets, Threading, AES Cryptography (Fernet), SQLite

- Constructed low-level TCP client-server chat server with custom length-prefix message framing and non-blocking sockets.
- Integrated client-side AES symmetric encryption ensuring zero plaintext storage or relay on central host.

RESEARCH & PUBLICATIONS

“Revolutionizing Intelligent Food Packaging: Nanomaterials, AI-ML, and Their Seamless Integration”
Published in **Food Reviews International (Taylor & Francis)** • Vol. 42, No. 6 •
DOI: 10.1080/87559129.2026.2631781 ↗

ACHIEVEMENTS & CERTIFICATIONS

- **400+ LeetCode DSA Problems:** Solved 400+ problems in C++ covering Binary Search, DP, Graphs, and Backtracking (Profile ↗).
- **Smart India Hackathon (SIH):** Cleared BPIT internal college rounds in both 2nd and 3rd year with *MineGov AI* for Ministry of Coal / Coal India (final round results awaited).
- **NPTEL IIT Kharagpur – Machine Learning:** *Elite + Silver* Certification with **83% consolidated score**.
- **Generative AI Studio:** Certified in prompt engineering, embeddings, and foundational LLM integrations (2025).
- **Web Development Internship:** Completed frontend internship at IAmigos Foundation (IAF).